Twiggle: Turning shoppers into buyers with Google Cloud

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Twiggle built a sc flexible platform, security standarc easily with Comp keeping latency t minimum and se premium.

Google Cloud results

- Enables smooth, fast service

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About Twiggle

Founded in 2014, Twiggle brings the power of human understanding to the ecommerce experience through AI-powered retail solutions that better connect shoppers to the products retailers sell online. Using the most advanced technologies in Natural Language Processing (NLP), machine learning, and knowledge engineering, Twiggle aims to facilitate a digital shopping experience that feels like the best in-store experience.

across the world with Compute Engine instances running on Google's multi-region network

- Helps maintain a high level of security, while simplifying permissions and access admin with Cloud Identity and Access Management
- Customizes each instance according to the task at hand while keeping costs low

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Industries: Retail & Consumer Goods, Technology

Location: Israel

(https://cloud.google.com/load-balancing/),

Google Cloud (https://cloud.google.com/)

Dataflow (https://cloud.google.com/dataflow/)

Cloud Dataproc (https://cloud.google.com/dataproc/)

Cloud Datalab (https://cloud.google.com/datalab/)

Cloud AI & Machine Learning

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud Memorystore (https://cloud.google.com/memorystore/)

Cloud Storage (https://cloud.google.com/storage/)

Compute Engine (https://cloud.google.com/compute/)

Cloud IAM (https://cloud.google.com/iam/)

TensorFlow (https://www.tensorflow.org/)

Cloud AI & Machine Learning (https://cloud.google.com/products/ai/)

In the world of ecommerce, competition is tough and margins are tight. Every sale counts, so when shoppers visit websites, search for items, and then don't buy anything, sellers have to ask themselves why. Twiggle (https://twiggle.com/) brings the power of human understanding to the ecommerce experience to help retailers around the world better connect shoppers to the products they're looking to buy.

Twiggle's AI-powered search, catalog enrichment, and analytics solutions enable retailers to enhance the search and discovery experience, optimize their product data, and make smarter merchandising decisions. Twiggle turns shoppers into buyers by leveraging its proprietary universal product ontology and Natural Language Analyzer, combined with a deep pool of knowledge engineers. With a deeper understanding of user intent and behavior, as well as product data, retailers can provide a frictionless ecommerce experience that drives conversions and inspires brand loyalty.

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Twiggle's solutions are based around Natural Language Processing (NLP) rather than key words. Twiggle understands, for instance, that searching for "a dress for a wedding" is not the same as searching for "a wedding dress" and will filter the results accordingly. Clients can connect their search engines to Twiggle via an API, providing the enhancement without the need to disrupt or modify any important infrastructure components. By 2017, as the company matured, Twiggle decided to overhaul its cloud infrastructure to help ensure that it had a solution that would build a strong foundation for the future and work in the best interests of its growing customer base which includes top multinational enterprises such as Walmart and Shopstyle. To do that it turned to Google Cloud (https://cloud.google.com/).

"For our product to work, our infrastructure has to be flexible, highly secure and, very importantly, it has to have very low latency. Every millisecond counts when it comes to search conversions," says Amir Di-Nur, VP of Research and Development at Twiggle. "The solution that best fit our needs and those of our customers was Google Cloud."

Maximum flexibility with Google Compute Engine

When Twiggle launched in 2014, and started building out the development and production environments for its product, its team did not have the time to learn new technologies and used the technology and cloud provider with which they were already familiar. "Like any startup in the early stages, speed of development was key for us," says Amir. By the time the product had come to market, Twiggle's customers had suggested the company change its cloud provider to one that was more suited for the ecommerce market. Twiggle's challenge was to find a cloud vendor that could suit its customers' preferences while fulfilling the service requirements of the existing solution. After assessing the major cloud providers capable of providing the scale of service Twiggle required, the company chose Google Cloud for the strength of its security, its compatibility with continuous integration/continuous development (CI/CD) and open source methods, and its multi-region network, which could deliver top-quality service to clients around the world.

In 2017, Twiggle started working with Google Cloud in Israel to find a new solution. The focus was initially on the production environment. This was the most important for customers and the easiest to migrate, as it was built as much as possible with open source, non-proprietary tools. Working closely with the Google Cloud services team, Twiggle built the new infrastructure for its production environment in just five weeks without any disruption to its customers. "The service and help we received from Google was excellent," says Amir.

"In our production environment we need to be flexible. Some of our services run all the time, and some run once a day or once a week. Google Compute Engine made it easy to run the services we wanted, whenever we wanted."

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Compute Engine (https://cloud.google.com/compute/) formed the backbone of Twiggle's new production environment. With the new architecture came a focus on services and projects, as opposed to hardware demands, which required Twiggle's DevOps team to think in a new way about assigning resources. With the ability to spin up and shut down servers on demand and an easy to use console panel, Twiggle found Compute Engine more than capable of handling the company's complex service requirements.

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In addition to Compute Engine, Twiggle used Cloud Storage (https://cloud.google.com/storage/) to hold its static assets and Cloud Load Balancing (https://cloud.google.com/load-balancing/) to help ensure traffic stayed consistent and smooth for its customers. The company also used managed REDIS services with Cloud Memorystore (https://cloud.google.com/memorystore/), keeping latency to a minimum.

For its development environment, which was much more reliant on proprietary tools than the open source production environment, Twiggle took a very careful and considered approach. Piece by piece, the company worked closely with Google to find the right products and architecture to replace the previous environment, with plans to complete the development migration in early 2019. So far, Twiggle has been making use of Cloud Dataflow (https://cloud.google.com/dataflow/) and Cloud Dataproc (https://cloud.google.com/dataproc/) to enhance its data pipelining capabilities, while its data scientists have been working with Cloud Datalab (https://cloud.google.com/datalab/) to run quick analyses and look for useful insight.

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Safer, simpler security protocols for peace of mind

With Google Cloud, Twiggle has built an infrastructure that delivers a smooth, high-speed service to its customers wherever they are in the world, thanks to Google's global network. At the same time, Google's expertise and reputation for strong security, has helped ease customer concerns about the security of their data. "Things like data encryption being the default for storage or Google's compliance with the SOC 2 (https://cloud.google.com/security/compliance/soc-2/) recommendations make security much easier for us than before," says Amir.

Cloud Identity and Access Management (https://cloud.google.com/iam/) helps maintain the security of Twiggle's work, while simplifying permissions and access management. For the IT department, this means far less time spent on administrative tasks and more time improving the product and developing new features. Meanwhile, Twiggle managed to reduce its infrastructure costs and expand its capabilities.

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With the migration of the development environment over to Google Cloud almost complete, Twiggle is already looking ahead to what other tools it might use in the future. The company has been experimenting with Kubernetes (https://cloud.google.com/kubernetes/) and containerized applications as a way of providing consistency of service and keeping latency down. Twiggle has also been exploring TensorFlow (https://www.tensorflow.org/) and the Cloud Machine Learning (https://cloud.google.com/products/ai/) tools to see what it can add to and optimize its own AI technology.

"The Google Cloud team here in Israel has been great during the transition and also in helping us think about the architecture and solutions that will suit us best," says Amir.

Learn More

Reach out to our team to see how Google Cloud can help your business

https://cloud.google.com